INCH-POUND

MIL-I-36783A
23 June 1995
SUPERSEDING
MIL-I-36783
25 July 1972

MILITARY SPECIFICATION

INSERT, OPTICAL, CHEMICAL BIOLOGICAL MASK, UNIVERSAL SIZE

This specification is approved for use by the Defense Personnel Support Center, the Directorate of Medical Materiel, DPSC-MA, 2800 S. 20th Street, Philadelphia, PA 19145, Department of the Defense and is available for use by all Departments and Agencies of the Department of Defense.

1. Scope.

1.1 This specification covers an optical insert for use with protective masks (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-MA, 2800 South 20th Street, Philadelphia, PA 19145, by using the self-addressed Standardization Document Improvement Proposal, (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6540

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SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129-2 - Marking for Shipment and Storage - Medical Material.

(Unless otherwise indicated, copies of federal and military specifications, standards, and publications are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the specification.

DRAWING

Defense Personnel Support Center (DPSC) Drawing No. 24138
Insert, Optical, CBR Mask, Universal Size.

(Copies of the drawing can be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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ASTM STANDARDS

American Society for Testing and Materials (ASTM)

ASTM D638	Test for Tensile Properties of Plastics.
ASTM B206	Copper-Nickel-Zinc Alloy Wire and Copper-Nickel Alloy Wire.
ASTM D2240	Rubber Property-Durometer Hardness.
ASTM D1238L	Measuring Flow Rates of Thermoplastics by Extrusion Plastometer.
ASTM D1505	Test for Density of Plastics by the Density-Gradient Technique.
ASTM D1525	Test for Vicat Softening Point of Plastics.
ASTM D1248	Specification for Polyethylene Plastic Molding and Extrusion Materials.
ASTM D1974	Standard Practice for Methods of Closing, Sealing, and Reinforcing Fiberboard Shipping Containers.
ASTM D3951	Standard Practice for Commercial Packaging.
ASTM D5118	Standard Practice for Fabrication of Fiberboard Shipping Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137).

ANSI/ASQC Z1.4 Sampling Procedures and Tables for Inspection of Attributes

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd St., New York, NY 10036-8002)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Components. The optical insert shall consist of a polyethylene plastic front fitted with metal temporal mounting struts.

3.2 Materials. Materials shall conform to the applicable portions of the specifications, standards, and drawing specified herein and shall be capable of meeting the requirements specified therein.

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3.2.1 Plastic. The plastic front shall be molded from medium density linear polyethylene plastic, conforming to American Society for Testing and Materials (ASTM) Standard D1248-II-A3. Plastic shall conform to the physical properties shown in table I, when tested as specified in section 4.

Table I. Physical Property Requirements

Property	Requirement
Density (grams per cubic centimeter)	0.926 to 0.940
Softening point (Vicat, ^{o}C.), minimum	110
Tensile strength at yield point, (pounds per square inch) minimum	2700
Flow rate, grams per 10 minutes, maximum	5.0
Hardness, Shore D scale	55 to 60

3.2.2 Temporal strut. Temporal strut shall be nickel silver conforming to copper alloy UNS number C74500 of ASTM B206.

3.2.2.1 Tensile strength. The strut shall be annealed to a tensile strength of 73,000 pounds per square inch.

3.3 Style, design, and construction. The optical insert shall conform to the style, design, and construction shown in Defense Personnel Support Center Drawing 24138. Front corners of the end sections shall be rounded as shown on the drawing.

3.3.1 Dimensions. Dimensions of the insert shall be as shown on the drawing with the tolerances specified.

3.3.2 Front. The front shall be molded with the integral plastic hinge as shown on the drawing and shall contain a 90 degree angle lens groove in the eyewire.

3.3.3 Temporal strut. The temporal strut shall be 0.078 inch diameter rod or wire and shall be imbedded to the full depth of the temporal slot as shown on the drawing.

3.4 Color. The insert front color shall be the color of the natural polymer, i.e. no pigments or dyes added.

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3.5 Lens insertion and retention. The flexibility and shape retention properties of the plastic front shall permit insertion of properly finished, cut and beveled lenses. The lenses shall be properly retained in the eyewire groove without danger of unintentional ejection.

3.6 Eyewire sample tests. The finish insert sample shall be tested as described in 4.4.5.1 Expansion test, 4.4.5.2 Bridge flexure and Tensile test, 4.4.5.3 Compression test and 4.4.5.4 Twisting test. Neither fracture or break shall occur.

3.7 Identification marking and labeling. The optical inserts shall be legibly and permanently marked on the inside of the front on the left or right side of the hinge with the manufacturer's name or registered trademark. The designation "66 PD" shall be marked as shown on the drawing.

3.8 Workmanship. Optical inserts shall be free from defects which detract from their appearance or may impair their serviceability. The plastic front shall be smooth and free of foreign matter. Flashing and parting lines shall be held to a minimum. The temporal struts shall be free of burrs and sharp edges, and ends shall be smooth and rounded.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determined by technical means of physical and chemical properties) of the item.

4.1.4 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination Sampling for examination shall be conducted in accordance with ANSI Z1.4 and table II. The sample unit for examination shall be one unit of issue (one insert).

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Table II. Sampling and Acceptance Numbers for Examination

	Inspection level	AQL (percent defective)
For visual examination		
Major A	II	1.0
Major B	S-3	1.0
Minor A	II	4.0
Minor B	S-3	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with ANSI Z1.4, inspection level S-1, except that the acceptance number shall be zero for all sample sizes and a minimum of three samples shall be taken.

4.3. Examination. The optical insert shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the following classification of defects. Additional defects shall be classified in accordance with ANSI Z1.4.

Categories and Defects

Major A
- 101 Eyewire groove not within tolerance.
- 102 Hinge damaged or not properly molded.
- 103 Dimensions of front not within tolerance (PD, eyewire, groove, hinge).
- 104 Cracks in plastic front.

Major B
- 151 Diameter of temporal strut not within tolerances.
- 152 Struts not properly embedded in slots.
- 153 Tabs missing from struts.
- 154 Struts not bent at proper angle.

Minor A
- 201 Metal parts not free of burrs, sharp edges and marks.
- 202 Plastic material not free of dirt, metal particles, or other foreign matter.

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203	Front not marked, improperly marked or illegibly marked.
204	Eyewire, groove, irregular.
205	Plastic front not free of scratches, surface irregularities and excessive flash and parting lines
206	Dimensions of end piece not within tolerances.
207	Strut shorter than specified.

Minor B

251	Strut longer than specified.
252	Slot dimensions not within tolerances.
253	Strut tabs not within tolerances.
254	Struts tabs stuck in wrong place.

4.4. Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis of metal. Four ounces of each lot of the metal used in construction of the temporal struts shall be selected for chemical analysis.

4.4.2 Lens insertion and retention. Insertion and retention of ophthalmic lenses shall be evaluated in accordance with 3.5.

4.4.3 Physical properties of polyethylene. The polyethylene shall be tested for conformance to requirements of table I, utilizing test procedures shown in table III

Table III. Test Methods

Property	ASTM Test Method
Density	D1505
Softening point	D1525
Tensile Strength	D638
Flow rate	D1238L
Hardness	D785

4.4.4 Metal tests. The chemical analysis and tensile tests shall be performed in accordance with referenced component specification, ASTM B206.

4.4.5 Eyewire strength and durability test. The finished insert sample shall be subjected to the following tests. All tests shall be at ambient conditions of 73°F and 50% RH. Test samples shall not be heated, cooled and subjected to mechanical or chemical treatment prior to testing.

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4.4.5.1 Expansion test. Use a flat aluminum gage, with chamfered edges, measuring 40 mm wide. Gage to be inserted into eyewire and twisted 3 revolutions to stretch the eyewire.

4.4.5.2 Bridge flexure and tensile test. Bridge shall be able to withstand 2000 flexes without complete fracture of any unit tested. Bridge should also be capable of witnstanding a straight pull of a minimum of 30 lbs. for 30 seconds.

4.4.5.3 Compression test. The insert shall be compressed to 80% of its original height for 3 minutes.

4.4.5.4 Twisting test. The insert shall be securely clamped, front face down with 1.0 inch hanging over the end of a chamfered corner (desk top) with the struts protruding upward, a sufficient force shall be applied vertically to the top of the strut until the strut has achieved an angle of 10 degrees with the horizontal.

4.5 Regulatory requirements.

4.5.1 Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statues, ordinances, and regulations.

4.5 2 Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

4.6 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129-2.

5. Packaging

5.1 Preservation. Preservation shall be level A, or C, as specified (see 6.1).

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5.1 1 Unit. Each (EA). One optical insert, as specified, constitutes one unit.

5.1.2.1 Level A.

5.1.2.1 Cleaning and drying. Optical inserts shall be cleaned and dried in accordance with MIL-P-116.

5.1.2 2 Preservation applications. Preservatives shall not be used.

5.1.2.3 Unit. One optical insert shall be packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2.4 Intermediate package. Twelve units shall be packaged in a box conforming to PPP-B-566, PPP-B-676 or ASTM D5118, class domestic. Closure shall be as specified in the applicable specification, except closure of containers conforming to ASTM D5118 shall be as specified in ASTM D1974.

5.1.3 Level C.

5.1.3.1 Unit package. Preservation for optical inserts shall conform to the requirements of ASTM D3951

5.1.3.2 Intermediate package. Twelve units shall be packaged in accordance with the requirements of ASTM D3951.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and conforming to ASTM D5118, class weather resistant. Closure, strapping shall be as specified in ASTM D1974. Each fiberboard box shall be water-resistant as specified in ASTM D1974.

5.2.2 Level B. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and conforming to ASTM D5118, class domestic. Closure shall be as specified in ASTM D1974.

5.2.3 Level C. Twenty-four intermediate packages (288 units) shall be packed in accordance with the requirement of ASTM D3951.

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5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order shall be used whenever total quantities for shipment to one destination equals 20 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147, on Type IV or Type V 4-way entry pallets. Pallet shall have a length of 40 inches and a width of 48 inches. Pallet loads, including the pallet, shall not exceed 54 inches in height, 43 inches in length and 52 inches in width.

5.2.5.1 Levels A and B. Optical inserts packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5 2 Level C. Optical inserts packed as specified in 5.2.3 shall be unitized as specified in ASTM D3951.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129-2.

5.4 General.

5.4 1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.

6. NOTES

6.1 Intended use. An optical insert for use with protective masks to position ophthalmic prescription lenses.

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6.2 Ordering data.

(a) Title, number, and date of the specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(c) National Stock Number.

6.3 Inserts to be used with Mask Model Nos. M14A2, M17A1, and M25.

6.4 Subject term (keyword) listing.

Insert
Optical insert
Gasmask insert
Goggles
Mask insert
Plastic insert

6.5 National Stock Number (NSN) Coverage. This specification covers the following National Stock Number:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6540-00-935-6573	INSERT, OPTICAL, CHEMICAL BIOLOGICAL MASK, UNIVERSAL SIZE.

MILITARY INTERESTS:

Custodians

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6540-1267
Location:\finals\S-36783A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-36783A	2. DOCUMENT DATE (YYMMDD) 950623

3. DOCUMENT TITLE
INSERT, OPTICAL, CHEMICAL BIOLOGICAL MASK, UNIVERSAL SIZE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Personnel Support Center ATTN DPSC-MAP	b. TELEPHONE (Include Area Code) (1) Commercial: (215) 737-8431 (2) AUTOVON: 444-8431
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church VA 22041 3466 Telephone (703) 756 2340 AUTOVON 289 2340

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